Signal Propagation (Mobile Apps/RF Clouds)

Quick briefing

Propagation models estimate radio coverage and link performance. Compare the modeled results on consistent assumptions, then verify the links and gaps that matter to the exercise.

Know the facts

Model inputs · Terrain, antenna height above local ground, transmit power, feed-line loss and receiver threshold affect the prediction.

Path profile · A path profile shows terrain relative to the direct path and its Fresnel zone. Diffraction can contribute to a usable signal even with an obstruction.

Link margin · Margin expresses the difference between predicted receive power and the required threshold. A small positive margin gives little room for worsening conditions.

Coverage and cellular tools · An area plot estimates the combined or individual coverage footprint. CellMapper can suggest tower locations and CloudRF predicts coverage from supplied inputs.

Relay and validation · A relay must support both legs of the intended connection. Confirm access, check the links in authorized field tests and update the model with the results.

Understand the reasoning

A stronger coverage color away from the required route can make a relay look more valuable than it is. Compare the corridor and endpoints the exercise depends on.

Keep the recorded model settings with the plot. Comparing different receiver thresholds or extents can make a model change look like a service improvement.

Philippine Army classroom exercise

A Philippine Army disaster-response exercise models two relay options for a supply route. One adds a larger colored area but leaves a gap near a delivery point. The other covers the delivery point, although one of its modeled link margins is small and crew access is uncertain.

Decision · Which option merits the first validation, and what would change that choice?

Worked analysis · Validate the second option first because it serves the delivery point, provided the crew can safely reach the site. Check both relay legs and the marginal link under representative conditions. If access or link performance fails, keep the approved alternate available and compare another option against the same route requirement.

How to practice

1 · Mark the exercise route, key contact points and endpoints on the same analysis map.

2 · Record the assumptions for the baseline and hold comparable settings constant when modeling a relay.

3 · Read both relay path profiles and margins, then compare the before-and-after coverage along the required route.

4 · Check access and conduct authorized link tests at the key points before updating the coverage plot and communications plan.

Check yourself

What is link margin in a model output? · Predicted received power minus the threshold needed for that link.

Why should a proposed relay be assessed on both sides? · The intended connection depends on usable paths from each endpoint to the relay.

Course source

Signal Propagation · Worked Coverage Study